Catherine Yeh

CS PhD Student @ Harvard University

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Education

Harvard University — Boston, MA
Ph.D. in Computer Science with M.S. in Passing — 2022 - present
Advisors: Martin Wattenberg, Fernanda Viégas, Hanspeter Pfister
Research Interests: Data Visualization, Human-AI Interaction, Interpretability

Williams College — Williamstown, MA
B.A. in Computer Science + Cognitive Science — 2018 - 2022
Advisor: Iris Howley
Thesis: Toward an Empirical Framework for Post-hoc Explainable AI
Graduated Summa Cum Laude (top 2% of class) with Highest Honors in CS

Research Experience

Industry

Google DeepMind, Inception Team — Mountain View, CA
Student Researcher — Summer 2026
Mentors: Peggy Chi, Lei Shi
Developed new interaction techniques for expressing and steering design intent when vibe coding.

Adobe Research, STORIE Lab — Seattle, WA
Research Scientist Intern — Summer 2025
Mentors: Bryan Wang, Anh Truong, Mira Dontcheva
Designed a video authoring tool to help creators craft compelling video stories by blending captured and generative media.

Apple Research, Human-Centered Machine Intelligence Group — Seattle, WA
Research Intern — Summer 2024
Mentors: Fred Hohman, Donghao Ren, Yannick Assogba, Dominik Moritz
Developed a visual, human-in-the-loop data augmentation tool for increasing data diversity in text datasets.

Microsoft Research, Human Understanding + Empathy Group — Redmond, WA
Research Intern — Summer 2023
Mentors: Gonzalo Ramos, Rachel Ng
Designed implicit and explicit feedback mechanisms for personalizing AI-powered writing experiences.

Adobe Research, Media Intelligence Lab — San Jose, CA
Research Scientist Intern — Summer 2022
Mentors: Franck Dernoncourt, Nedim Lipka
Developed exploratory prototypes showcasing vision for the next-gen document reader.

Microsoft Research, Productivity + Intelligence Group — Redmond, WA
Research Intern — Summer 2021
Mentors: Jenna Butler, Christian Bird
Built a self-reflection chatbot and personalized dashboards for studying hybrid workforce productivity and well-being.

Academic

Harvard University, Insight + Interaction Lab, Visual Computing Group — Boston, MA
Graduate Research Assistant — 2022 - present
Advisors: Martin Wattenberg, Fernanda Viégas, Hanspeter Pfister

Designing visualization-driven interfaces and techniques to support model interpretability and enable more productive, safe, and creative human-AI interactions.

Williams College, Computation^2 Lab — Williamstown, MA
Undergraduate Research Assistant — 2021-2022
Advisor: Molly Feldman
Analyzed and developed guidelines for replicability in human-computer interaction research.

Williams College, Human-AI Interaction Lab — Williamstown, MA
Undergraduate Research Assistant — 2019-2022
Advisor: Iris Howley
Studied knowledge tracing algorithms and developed an empirical framework for explainable AI informed by learning theory.

Williams College, Concepts + Categories Lab — Williamstown, MA
Undergraduate Research Assistant — 2018-2019
Advisor: Safa Zaki
Designed eye-tracking experiments to study cognitive learning strategies in categorization tasks.

Publications

* indicates equal contribution | Google Scholar

Conference

C1 **Vidmento: Creating Video Stories Through Context-Aware Expansion With Generative Video**
Catherine Yeh, Anh Truong, Mira Dontcheva, Bryan Wang
ACM Conference on Human Factors in Computing Systems (CHI), 2026
Paper

C2 **Story Ribbons: Reimagining Storyline Visualizations With Large Language Models**
Catherine Yeh, Tara Menon, Robin Singh Arya, Helen He, Moira Weigel, Fernanda Viégas, Martin Wattenberg
IEEE Visualization Conference (VIS), 2025
Paper · 🏆 Best Paper Honorable Mention (top 3%)

C3 **Chronotome: Real-Time Topic Modeling for Streaming Embedding Spaces**
Matte Lim*, **Catherine Yeh***, Martin Wattenberg, Fernanda Viégas, Panagiotis Michalatos
IEEE Visualization Conference (VIS), 2025
Paper · 🏆 Best Short Paper Honorable Mention (top 2.5%)

C4 **Exploring Empty Spaces: Human-in-the-Loop Data Augmentation**
Catherine Yeh, Donghao Ren, Yannick Assogba, Dominik Moritz, Fred Hohman
ACM Conference on Human Factors in Computing Systems (CHI), 2025
Paper · 🏆 Best Paper Honorable Mention (top 5%)

C5 **AttentionViz: A Global View of Transformer Attention**
Catherine Yeh, Yida Chen, Aoyu Wu, Cynthia Chen, Fernanda Viégas, Martin Wattenberg
IEEE Visualization Conference (VIS), 2023
Paper

C6 **Designing for Student Understanding of Learning Analytics Algorithms**
Catherine Yeh, Noah Cowit, Iris Howley
International Conference on Artificial Intelligence in Education (AIED), 2023
Paper

Workshop

W1 **What Kind of User Are You? Uncovering User Models in LLM Chatbots**
Yida Chen, Aoyu Wu, Trevor DePodesta, **Catherine Yeh,** Lena Armstrong, Kenneth Li, Nicholas Castillo Marin, Oam Patel, Jan Riecke, Shivam Raval, Olivia Seow, Martin Wattenberg, Fernanda Viégas
ICML Actionable Interpretability Workshop (AIW), 2025
Paper

W2 **Envisioning the Next-Gen Document Reader**
Catherine Yeh, Franck Dernoncourt, Nedim Lipka

AAAI Workshop on Scientific Document Understanding (SDU), 2023
Paper

W3 **Tests, Memory, and Artificial Intelligence: How Can We Know What People Know?**
Noah Cowit, **Catherine Yeh,** Iris Howley
IEEE VIS Workshop on Visualization for AI Explainability (VISxAI), 2019

Other

O1 **Walk a Mile in Their Shoes: The Covid Pandemic Through the Lens of Four Tech Workers**
Jenna Butler, **Catherine Yeh**
Communications of the ACM, 2022. Previously also published in ACM Queue.
Paper

Preprint / In Submission

P1 **VideoTrail: Interactive Video Foraging Through Hierarchical Visualization**
Catherine Yeh, Bryan Wang, Fernanda Viégas, Martin Wattenberg

P2 **Neural Representations As New Modes Of Interactivity**
Stanley Huang, Anthony Baez, **Catherine Yeh,** Pat Pataranutaporn

P3 **GhostWriter: Augmenting Collaborative Human-AI Writing Experiences Through Personalization and Agency**
Catherine Yeh, Gonzalo Ramos, Rachel Ng, Andy Huntington, Richard Banks
Paper

Patents

Video Sequencing and Gap Detection for Authoring Videos Including Captured and Generated Content
Application Filed December 2025

Generative Neural Network Model Style Guide Management
US Patent App. 18400715

Invited Talks + Panels

Research

Telling Stories With and About AI: Visualizing and Shaping Meaning from Unstructured Data
Kempner Institute, April 2026

Exploring Empty Spaces: Human-in-the-Loop Data Augmentation
Apple Research Machine Learning Summit, March 2025

Story Ribbons: Reimagining Storyline Visualizations With Large Language Models
Center for Human-Driven AI Research and Methods (CHARM) @ Harvard University, February 2026
Kempner Institute, February 2025

Other

Crash Course on Securing PhD Research Internships
Harvard University Computer Science Department, November 2025

Principles of Data Visualization
Carnegie Mellon University Language Technologies Institute (LTI), June 2025

Panel on Effective Research Practices for PhD Students
Harvard University Computer Science Department, February 2025

Honors + Awards

2025 **Best Paper Honorable Mention (x2),** IEEE Visualization Conference
Best Paper Honorable Mention, ACM Conference on Human Factors in Computing Systems
Graduate Collegiate Award Finalist, National Center for Women & Information Technology

10 Under 10 Outstanding Alumni Award, The Overlake School

2024 **Graduate Research Fellowship,** National Science Foundation
Graduate Fellowship, Kempner Institute

2022 **Sigma Xi International Scientific Research Honor Society,** Williams College
Outstanding Undergraduate Researcher Award Runner Up, Computing Research Association

2021 **Undergraduate Collegiate Award Finalist,** National Center for Women & Information Technology
Women-in-Technology Scholarship Finalist, Adobe Research
Phi Beta Kappa National Honor Society (top 5% of class), Williams College
ACM Student Research Competition Finalist, Grace Hopper Conference
Ward Prize Finalist, Williams College

2019 **Computer Science Class of 1960s Scholar (2019 - 2022),** Williams College
3rd Place Winner, National Undergraduate Statistics Project Competition
Summer Science Research Fellowship, Williams College
Scholarship Recipient, Grace Hopper Conference

2018 **Dean's List (2018 - 2022),** Williams College

Other Industry Experience

Oracle Cloud Infrastructure — Seattle, WA
Software Engineer Intern — Summer 2020
Built dashboard and visual database for regional compliance programs.

Sunshine — Palo Alto, CA
Software Engineer Intern — Winter 2020
Created dashboard for AI-driven consumer app.

Teaching

Harvard University

CS 279R: Research Topics in Human-Computer Interaction - Fall 2023

Williams College

CSCI 334: Principles of Programming Languages - Spring 2021, Fall 2021, Spring 2022
CSCI 315: Computational Biology - Fall 2021
CSCI 256: Algorithm Design + Analysis - Spring 2020
CSCI 136: Data Structures + Advanced Programming - Spring 2019, Fall 2019, Fall 2020

Mentoring

* indicates mentees who co-authored publications

Research

Allison Huang, University of Southern California '26
Generative UIs for Verifying LLM Reasoning, Summer 2026

Helen He*, Harvard University '26
LLM-Augmented Storyline Visualizations, Spring 2025

Vicki Xu, Harvard University '23
Logit Lens on Vision Transformer, Fall 2023

Cynthia Chen*, Harvard University '24
Visualizations for Transformer Interpretability, Spring 2023

Other

2025 - present **VISBuddies Mentor,** IEEE VIS
2024 - present **SEAS Research Program Mentor,** Harvard University
2022 - present **Women in STEM Mentor,** Harvard University
2020 - 2022 **Underrepresented Identities in CS Mentor,** Williams College

Service

Organizer

IEEE VIS Workshop on Visualization for AI Explainability (VISxAI) - 2025, 2026
IEEE VIS Workshop on Visualization for GenAI, Agents, and the Future of VIS (VISxGenAI) - 2025

Program Committee

IEEE VIS Workshop on Visualization for GenAI, Agents, and the Future of VIS (VISxGenAI) - 2026
IEEE VIS Workshop on Visualization for Communication (VisComm) - 2025
IEEE VIS Workshop on Visualization for AI Explainability (VISxAI) - 2024

Reviewer

* indicates outstanding review recognitions

Conference

ACM Symposium on User Interface Software and Technology (UIST) - 2026*
Eurographics Conference on Visualization (EuroVis) - 2026*
IEEE Pacific Visualization Conference (PacificVis) - 2026
IEEE Visualization Conference (IEEE VIS) - 2025
ACM Conference on Human Factors in Computing Systems (CHI) – 2025*, 2026*, 2027
ACM Conference on Designing Interactive Systems (DIS) – 2024

Journal

Journal of Visualization and Interaction (JoVI) – 2025
ACM Transactions on Intelligent Systems and Technology (TIST) – 2024

Volunteer

IEEE VIS Student Volunteer - 2023

Misc

Art Portfolio